

Matrix – Determinant – System of Equations

Determinant

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1 Determinant

2 Applications



Definition (Determinant)

Let $A \in M_n(K)$. Determinant of A denoted as $\det(A)$ is a real or complex number defined as follows.

- $n = 1 : A = [a_{11}] \implies \det(A) = |a_{11}| = a_{11}.$
- $n = 2 : A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} : \det(A) = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}.$
- $n \geq 2 : A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}_n \implies \det(A) = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n},$

where

$C_{ij} = (-1)^{i+j} \begin{vmatrix} \text{submatrix formed by deleting} \\ \text{the } i\text{-th row and } j\text{-th column of } A \end{vmatrix}$, is the (i,j) -**cofactor** of A



Example 1.

Let $A = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 3 & 0 \\ 3 & 2 & 4 \end{pmatrix}$. Compute $\det(A) = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 3 & 0 \\ 3 & 2 & 4 \end{vmatrix}$



Example 1.

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$$\det(A) = 1(-1)^{1+1} \begin{vmatrix} 3 & 0 \\ 2 & 4 \end{vmatrix} + 2(-1)^{1+2} \begin{vmatrix} 2 & 0 \\ 3 & 4 \end{vmatrix} - 3(-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = 12 - 16 + 15 = 11.$$



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→ cofactor expansion across the first row of A

Cofactor expansion down the j -th column

$$\det(A) = a_{k1}C_{k1} + a_{k2}C_{k2} + \cdots + a_{kn}C_{kn} = a_{1k}C_{1k} + a_{2k}C_{2k} + \cdots + a_{nk}C_{nk}$$

$$\det(A) = (-3)(-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} + 4(-1)^{3+3} \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 15 - 4 = 11.$$



Example 2.

$$\text{Compute } \det(A) = \begin{vmatrix} 1 & 2 & -1 & 2 \\ 2 & 1 & 3 & -1 \\ 0 & 0 & -3 & 0 \\ 2 & 1 & -1 & 0 \end{vmatrix}.$$



Example 2.

Compute $\det(A) = \begin{vmatrix} 1 & 2 & -1 & 2 \\ 2 & 1 & 3 & -1 \\ 0 & 0 & -3 & 0 \\ 2 & 1 & -1 & 0 \end{vmatrix}$.

Use the cofactor expansion across the third row

$$\det(A) = -3 \cdot (-1)^{3+3} \cdot \begin{vmatrix} 1 & 2 & 2 \\ 2 & 1 & -1 \\ 2 & 1 & 0 \end{vmatrix}$$

Use the cofactor expansion down the third column

$$= -3 \left(2(-1)^{1+3} \cdot \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} - 1(-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \right) = -3(0 - 3) = 9$$



Example 3. Compute

$$\text{a) } \begin{vmatrix} 1 & 2 & 0 \\ 3 & 2 & 0 \\ -1 & 1 & 0 \end{vmatrix}$$



Example 3. Compute

$$\text{a) } \begin{vmatrix} 1 & 2 & 0 \\ 3 & 2 & 0 \\ -1 & 1 & 0 \end{vmatrix} = 0 \quad \text{b) } \begin{vmatrix} 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & 4 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -4 \end{vmatrix}$$



Example 3. Compute

$$\text{a) } \begin{vmatrix} 1 & 2 & 0 \\ 3 & 2 & 0 \\ -1 & 1 & 0 \end{vmatrix} = 0 \quad \text{b) } \begin{vmatrix} 1 & 2 & 1 & -1 \\ 0 & 3 & 1 & 4 \\ 0 & 0 & -2 & 1 \\ 0 & 0 & 0 & -4 \end{vmatrix} = 1 \cdot 3 \cdot (-2) \cdot (-4) = 24.$$



Properties of Determinants. Let $A, B \in M_n$.

- i) $\det(A^T) = \det(A)$.
- ii) If any row or column is all zeros, $\det(A) = 0$.
- iii) If A is an upper or lower triangular matrix, $\det(A) = a_{11} \cdot a_{22} \cdot \dots \cdot a_{nn}$
- iv) Row/Column Operations:
 - If $A \xrightarrow{r_i \rightarrow \alpha r_i} B$, then $\det(B) = \alpha \det(A) \implies \det(\alpha A) = \alpha^n \det(A)$.
 - If $A \xrightarrow{r_i + \beta r_j} B$, then $\det(B) = \det(A)$.
 - If $A \xrightarrow{r_i \leftrightarrow r_j} B$, then $\det(B) = -\det(A)$.
- v) $\det(AB) = \det(A) \cdot \det(B)$
- vi) $\det(A^{-1}) = \frac{1}{\det(A)}$, $\det(A^m) = \det(A)^m, m \in \mathbb{Z}_*$.

Remark: In general, $\det(A + B) \neq \det(A) + \det(B)$.



Example 4.

Compute

$$a) \begin{vmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ 3 & 1 & 0 \end{vmatrix}, \quad b) \begin{vmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 5 & 0 \\ 3 & 2 & 6 & -2 \\ -2 & 1 & 3 & 1 \end{vmatrix}$$

$$c) \begin{vmatrix} 0 & 1 & -1 & 3 \\ 2 & 1 & 1 & -2 \\ 4 & 1 & 3 & 2 \\ -2 & 1 & 0 & 1 \end{vmatrix}, \quad d) \begin{vmatrix} m & 1 & 1 & 1 \\ 1 & m & 1 & 1 \\ 1 & 1 & m & 1 \\ 1 & 1 & 1 & m \end{vmatrix}$$



Example 4. Compute

$$\text{a) } \begin{vmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ 3 & 1 & 0 \end{vmatrix}$$



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$$\text{a) } \begin{vmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ 3 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & -5 & 3 \end{vmatrix} = 1 \cdot (-1)^{1+1} \cdot \begin{vmatrix} -1 & 1 \\ -5 & 3 \end{vmatrix} = 2.$$

$$\text{b) } \begin{vmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 5 & 0 \\ 3 & 2 & 6 & -2 \\ -2 & 1 & 3 & 1 \end{vmatrix}$$



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$$\text{a) } \begin{vmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ 3 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & -5 & 3 \end{vmatrix} = 1 \cdot (-1)^{1+1} \cdot \begin{vmatrix} -1 & 1 \\ -5 & 3 \end{vmatrix} = 2.$$

$$\text{b) } \begin{vmatrix} 1 & 1 & 2 & -1 \\ 2 & 3 & 5 & 0 \\ 3 & 2 & 6 & -2 \\ -2 & 1 & 3 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 2 & -1 \\ 0 & 1 & 1 & 2 \\ 0 & -1 & 0 & 1 \\ 0 & 3 & 7 & -1 \end{vmatrix} = 1 \cdot (-1)^{1+1} \begin{vmatrix} 1 & 1 & 2 \\ -1 & 0 & 1 \\ 3 & 7 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 2 \\ -1 & 0 & 1 \\ -4 & 0 & -15 \end{vmatrix} = 1 \cdot (-1)^{1+2} \begin{vmatrix} -1 & 1 \\ -4 & -15 \end{vmatrix} = -1(15 + 4) = -19.$$



Example 5. Compute

$$\text{a) } \begin{vmatrix} 0 & 1 & -1 & 3 \\ 2 & 1 & 1 & -2 \\ 4 & 1 & 3 & 2 \\ -2 & 1 & 0 & 1 \end{vmatrix}$$

$$\text{b) } \begin{vmatrix} m & 1 & 1 & 1 \\ 1 & m & 1 & 1 \\ 1 & 1 & m & 1 \\ 1 & 1 & 1 & m \end{vmatrix}$$



Example 6.

Given $A, B \in M_3$ such that $\det(A) = 2, \det(B) = 3$. Compute $|2A^3|, |3AB^T|, \det(2A^T \cdot (3B^2)^{-1}), \det((2A^2)^{-1} \cdot (3B^T)^2)$.

a) $|2A^3|$.



Example 6.

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a) $|2A^3|.$

$$|2A^3| = 2^3 \cdot |A|^3 = 8 \cdot 2^3 = 64.$$

b) $|3AB^T|$



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Given $A, B \in M_3$ such that $\det(A) = 2, \det(B) = 3$. Compute $|2A^3|, |3AB^T|, \det(2A^T.(3B^2)^{-1}), \det((2A^2)^{-1}.(3B^T)^2)$.

a) $|2A^3|.$

$$|2A^3| = 2^3 \cdot |A|^3 = 8 \cdot 2^3 = 64.$$

b) $|3AB^T|$

$$|3AB^T| = 3^3 |A| |B| = 27 \cdot 2 \cdot 3 = 162.$$

c) $\det(2A^T.(3B^2)^{-1})$



Example 6.

Given $A, B \in M_3$ such that $\det(A) = 2, \det(B) = 3$. Compute $|2A^3|, |3AB^T|, \det(2A^T \cdot (3B^2)^{-1}), \det((2A^2)^{-1} \cdot (3B^T)^2)$.

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b) $|3AB^T|$

$$|3AB^T| = 3^3 |A| |B| = 27 \cdot 2 \cdot 3 = 162.$$

c) $\det(2A^T \cdot (3B^2)^{-1}) = 2^3 \cdot \det(A^T) \frac{1}{\det(3B^2)} = 8 \cdot 2 \cdot \frac{1}{3^3 \cdot \det(B)^2} = \frac{16}{27 \cdot 3^2} = \frac{16}{243}.$

d) $\det((2A^2)^{-1} \cdot (3B^T)^2) = \frac{1}{2^3 |A|^2} \cdot (3^3 \cdot |B|)^2 = \frac{6561}{32}.$



Invertible matrix

A square matrix A is invertible if and only if $\det(A) \neq 0$.



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Example 7.

Find m such that $A.B$ invertible. Given $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 2 \\ 0 & -1 & 3 \end{pmatrix}$, $B = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 1 & 1 \\ m & 2 & 1 \end{pmatrix}$.



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AB is invertible iff $\det(AB) \neq 0$

$$\iff \det(A) \cdot \det(B) \neq 0 \iff -1 \cdot (-4m - 2) \neq 0 \iff m \neq -\frac{1}{2}.$$



Example 8.

Given $A = \begin{pmatrix} 0 & 1 & -2 \\ 3 & 2 & -1 \\ 1 & m & 2 \end{pmatrix}$. Find m such that A is invertible.



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A is invertible iff

$$\det(A) \neq 0 \iff -6m - 3 \neq 0 \iff m \neq \frac{-1}{2}.$$



Theorem. An Inverse Formula

Let A be an invertible $n \times n$ matrix. Then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}A$$

where $\text{adj}A = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{pmatrix}$ is called the **adjugate** (or classical adjoint) of A .

The adjugate matrix is the transpose of the matrix of cofactors.



Example 9.

Find the inverse of the matrix $A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{pmatrix}$



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The nine cofactors are

$$C_{11} = + \begin{vmatrix} -1 & 1 \\ 4 & -2 \end{vmatrix} = -2, \quad C_{12} = - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = 3, \quad C_{13} = + \begin{vmatrix} 1 & -1 \\ 1 & 4 \end{vmatrix} = 5$$

$$C_{21} = - \begin{vmatrix} 1 & 3 \\ 4 & -2 \end{vmatrix} = 14, \quad C_{22} = + \begin{vmatrix} 2 & 3 \\ 1 & -2 \end{vmatrix} = -7, \quad C_{23} = - \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = -7$$

$$C_{31} = + \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 4, \quad C_{32} = - \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} = 1, \quad C_{33} = + \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3$$

$$\text{Thus } \text{adj}A = \begin{pmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{pmatrix}.$$



According to the Theorem,

$$(\text{adj}A)A = \begin{pmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{pmatrix} \begin{pmatrix} 2 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 4 & -2 \end{pmatrix} = \begin{pmatrix} 14 & 0 & 0 \\ 0 & 14 & 0 \\ 0 & 0 & 14 \end{pmatrix} = 14I$$

Therefore, $\det(A) = 14$ and we have

$$A^{-1} = \frac{1}{14} \begin{pmatrix} -2 & 14 & 4 \\ 3 & -7 & 1 \\ 5 & -7 & -3 \end{pmatrix}.$$



Example 10.

Find the degree of the polynomial

$$\text{a) } f(x) = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 1 & x \\ 3 & -5 & x^2 \end{vmatrix}$$



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$$\text{a) } f(x) = \begin{vmatrix} 1 & 2 & -3 \\ 2 & 1 & x \\ 3 & -5 & x^2 \end{vmatrix} = ax^2 + bx + c, a = \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = -3 \Rightarrow \deg(f(x)) = 2.$$

$$\text{b) } f(x) = \begin{vmatrix} 1 & x & x^2 & x^3 \\ 2 & 1 & -1 & 0 \\ 3 & 2 & 1 & 2 \\ 1 & 0 & -3 & 2 \end{vmatrix}$$



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$$\text{b) } f(x) = \begin{vmatrix} 1 & x & x^2 & x^3 \\ 2 & 1 & -1 & 0 \\ 3 & 2 & 1 & 2 \\ 1 & 0 & -3 & 2 \end{vmatrix} = ax^3 + bx^2 + cx + d, \text{ where}$$

$$a = \begin{vmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 1 & 0 & -3 \end{vmatrix} = 0, \quad b = \begin{vmatrix} 2 & 1 & 0 \\ 3 & 2 & 2 \\ 1 & 0 & 2 \end{vmatrix} = 4$$

Therefore, $\deg(f(x)) = 2$.



1 Determinant

2 Applications



Example 12 (Leslie's model)

Suppose the age of a female of a certain animal species is divided into three classes: Class I from 0 to 3 years; Class II from 3 to 6 years; Class III from 6 to 9 years. After every three years, in Class I, the females do not reproduce; in Class II, each female gives birth to an average of 4 other females; in Class III, each female gives birth to an average of 2 other females. About 40% of the females in Class I and 20% of the females in Class II survive through the three-year period.



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Let $X_0 = \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix}$ be the initial number of females in each class.

The number of females in each class after 3 years (one cycle) is

$$X_1 = \begin{pmatrix} 4b_0 + 2c_0 \\ 0.4a_0 \\ 0.2b_0 \end{pmatrix} = \begin{pmatrix} 0 & 4 & 2 \\ 0.4 & 0 & 0 \\ 0 & 0.2 & 0 \end{pmatrix} \begin{pmatrix} a_0 \\ b_0 \\ c_0 \end{pmatrix} := LX_0.$$



Similarly, $X_2 = LX_1, X_3 = LX_2 \dots$ In general,

Leslie matrix

$$X_{n+1} = L.X_n, \quad X_n = L^n X_0,$$

Leslie matrix: $L = \begin{pmatrix} 0 & 4 & 2 \\ 0.4 & 0 & 0 \\ 0 & 0.2 & 0 \end{pmatrix}.$

X_n is the number of females in each class after n cycle ($n \times 3$ years).



Example 12 (cont.)

Suppose initially there are 1,000 females in each class. Calculate the number of females in each class after 9 years.



Example 12 (cont.)

Suppose initially there are 1,000 females in each class. Calculate the number of females in each class after 9 years.

We have $X_0 = \begin{pmatrix} 1000 \\ 1000 \\ 1000 \end{pmatrix}$. after 9 years = 3 cycles, the number of females in each class is

$$X_3 = L^3 \cdot X_0 = \begin{pmatrix} 0 & 4 & 2 \\ 0.4 & 0 & 0 \\ 0 & 0.2 & 0 \end{pmatrix}^3 \begin{pmatrix} 1000 \\ 1000 \\ 1000 \end{pmatrix} =$$



Example 13 (Leslie's model)

Suppose the age of a female of a certain animal species is divided into three classes: Class I from 0 to 2 years; Class II from 2 to 4 years; Class III from 4 years and older. After every two years, in Class I, each female gives birth to an average of 0.4 other females; in Class II, each female gives birth to an average of 3 other females; in Class III, each female gives birth to an average of 1 other female. About 30% of the females in Class I, 50% of the females in Class II, and 10% of the females in Class III survive after each two-year period.

a/ Construct the Leslie matrix for the model



Example 13 (Leslie's model)

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a/ Construct the Leslie matrix for the model

$$L = \begin{pmatrix} 0.4 & 3 & 1 \\ 0.3 & 0 & 0 \\ 0 & 0.5 & 0.1 \end{pmatrix}$$



b/

Suppose initially there are 10,000 females in each class. How many females will there be in each class after 8 years?



b/

Suppose initially there are 10,000 females in each class. How many females will there be in each class after 8 years?

We have $X_0 = \begin{pmatrix} 10000 \\ 10000 \\ 10000 \end{pmatrix}$. After 8 years = 4 cycles, the number of females in each class is

$$X_4 = L^4 \cdot X_0 = \begin{pmatrix} 0.4 & 3 & 1 \\ 0.3 & 0 & 0 \\ 0 & 0.5 & 0.1 \end{pmatrix}^4 \begin{pmatrix} 10000 \\ 10000 \\ 10000 \end{pmatrix} =$$



Example 14.

In a population of a species, the females are divided into three age classes: 0–1 year, 1–2 years, and over 2 years. We consider a Leslie model representing the increase in the

female population. The transition matrix of the model is given by
$$\begin{bmatrix} 0 & 6 & 4 \\ 0.7 & 0 & 0 \\ 0 & 0.5 & 0.6 \end{bmatrix}.$$

a/ What is the survival rate of the third age class after one year?

b/ What is the survival rate of a newborn after three years?

c/ Suppose initially there are 500 newborns, 400 adults, and 200 old individuals. How many will there be in each class after four years?



Example 15(Markov model).

Regarding the population in a city, it was found that each year, 10% of the people living in the suburbs move to the city centre, and 5% of the people in the city centre move to the suburbs. Assume that the number of births, deaths, and people moving from other places is negligible.

a/ Simulate the problem using a Markov model.



Example 15(Markov model).

Regarding the population in a city, it was found that each year, 10% of the people living in the suburbs move to the city centre, and 5% of the people in the city centre move to the suburbs. Assume that the number of births, deaths, and people moving from other places is negligible.

a/ Simulate the problem using a Markov model.

Let $X = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$ denote the population in the suburbs and the city centre, respectively.

After one year,

$$X_1 = \begin{pmatrix} (1 - 0.1)x_0 + 0.05y_0 \\ 0.1x_0 + (1 - 0.05)y_0 \end{pmatrix} = \begin{pmatrix} 0.9 & 0.05 \\ 0.1 & 0.95 \end{pmatrix} \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} := MX_0.$$



Generally,

Markov matrix

$$X_{n+1} = MX_n, \quad X_n = M^n X_0.$$

$M = \begin{pmatrix} 0.9 & 0.05 \\ 0.1 & 0.95 \end{pmatrix}$ Markov matrix (also called the transition probability matrix).

X_n denotes the population of the two regions after n years.

Properties of a Markov matrix

- i/ The elements of a Markov matrix belong to $(0, 1)$.
- ii/ The sum of the elements in each column equals 1.
- iii/ The matrix $A - I$ is not invertible.
- iv/ The linear system $(A - I)X = 0$ has infinitely many solutions.



b/

Suppose initially the population of the suburbs is 2 million and the city centre is 5 million. Calculate the population of each region after 5 years



b/

Suppose initially the population of the suburbs is 2 million and the city centre is 5 million. Calculate the population of each region after 5 years

We have $X_0 = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$.

$$X_5 = M^5 \cdot X_0 = \begin{pmatrix} 0.9 & 0.05 \\ 0.1 & 0.95 \end{pmatrix}^5 \cdot \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 2.185 \\ 4.813 \end{pmatrix} \text{ (million).}$$



c/

Suppose the total population in the city is 7.5 million. It is known that the population of each region remains constant each year. Calculate the population of each region



c/

Suppose the total population in the city is 7.5 million. It is known that the population of each region remains constant each year. Calculate the population of each region

Suppose the population of each region is $X_* = \begin{pmatrix} x_* \\ y_* \end{pmatrix}$. We have

$$\begin{cases} X_* = MX_* \\ x_* + y_* = 7.5 \end{cases} \iff \begin{cases} 0.9x_* + 0.05y_* = x_* \\ 0.1x_* + 0.95y_* = y_* \\ x_* + y_* = 7.5 \end{cases} \iff \begin{cases} x_* = 2.5 \\ y_* = 5 \end{cases} \text{ (million).}$$



Example 16.

Suppose in a town there are two supermarkets, A and B. According to a survey, each month 10% of A's customers switch to buying from B, and 15% of B's customers switch to buying from A. (Assume no new customers appear, none stop shopping, and each customer shops at only one supermarket.)

- a) Find the Markov matrix for this problem.
- b) It is known that the number of customers at each supermarket remains constant each month and the total number of customers for both supermarkets is 5,000. Find the number of customers at each supermarket."



Example 17.

Suppose in 2021, the population of city X and its suburbs is 20,000 and 10,000 respectively. According to research, each year about 10% of the city's population moves to the suburbs and 15% of the suburban population moves into the city.

- Estimate the population of city X in 2023.
- Suppose the total population in the city is 30,000. Knowing that the population in the two regions remains constant, calculate the population of each region.



Example 18.

An agricultural cooperative specialises in growing three main crops: cassava, sweet potatoes, and kudzu roots. To ensure crop rotation, each year the cooperative transfers a certain proportion of households growing one crop to grow another. This transition is represented by the following Markov matrix: $\begin{pmatrix} 0.7 & 0.1 & 0.2 \\ 0.2 & 0.6 & 0 \\ 0.1 & 0.3 & 0.8 \end{pmatrix}$. Currently, the number of households growing cassava, sweet potatoes, and kudzu roots is 200, 250, and 100 respectively.

- How many households will be growing sweet potatoes after 3 years?
- How many households were growing sweet potatoes last year?
- Suppose the total number of households in the region remains constant at 10,000. Knowing that the number of households growing each crop does not change over time, calculate the number of households growing each crop.

THANK YOU FOR ATTENTION

